

.

Intrusive and non-intrusive model reduction

Today

Last time

- Model reduction in a nutshell
- Abstract formulation of our problem
- Thermal block

Today

- Full model and Galerkin
- Approximation

Announcements

- Homework 1 posted, due 16.10

Continuity and coercivity:

Def: For a fixed $\mu \in D$, the bilinear form a is continuous on $V \times V$, if there exists $f(\mu) < \infty$ such that

$$|a(w, v; \mu)| \leq f(\mu) \|w\|_V \|v\|_V \quad \forall w, v \in V$$

Denote the smallest possible constant

$$f(\mu) = \sup_{w \in V \setminus \{0\}} \sup_{v \in V \setminus \{0\}} \frac{|a(w, v; \mu)|}{\|w\|_V \|v\|_V}$$

Def For a fixed $\mu \in D$, the bilinear form a is coercive on V if there exists $\alpha(\mu) > 0$ such that

$$a(v, v; \mu) \geq \alpha(\mu) \|v\|_V^2 \quad \forall v \in V$$

The largest $\alpha(\mu)$ is

$$\alpha(\mu) = \inf_{v \in V \setminus \{0\}} \frac{a(v, v; \mu)}{\|v\|_V^2}$$

with coercivity have

$$v \neq 0 \Rightarrow a(v, v; \mu) > 0$$

Def The linear form is continuous if $\delta(\mu) > 0$ with

$$|f(v; \mu)| \leq \delta(\mu) \|v\|_V \quad \forall v \in V.$$

Space of all cont. linear forms over V is dual V'

$$f \in V', \quad \|f\|_{V'} = \sup_{v \in V \setminus \{0\}} \frac{|f(v)|}{\|v\|_V}$$

Have for $f \in V'$ that for any fixed $v \in V$ and $v \neq 0$ that

$$\frac{|f(v)|}{\|v\|_V} \leq \sup_{v \in V \setminus \{0\}} \frac{|f(v)|}{\|v\|_V} = \|f\|_{V'}$$

Obtain

$$|f(v)| \leq \|f\|_{V'} \|v\|_V$$

Properties of thermal block

↳ "Suitable" space V and

$$0 < \mu_{\min} \leq k(x; \mu) \leq \mu_{\max}$$

then bilinear form is continuous and coercive

$$|a(w, v; \mu)| \leq \mu_{\max} \|w\|_V \|v\|_V$$

$$a(v, v; \mu) \geq \mu_{\min} \|v\|_V^2$$

and the linear form is cont.

Well-posedness (Lax-Milgram): Let V be a real Hilbert space. Let $a: V \times V \rightarrow \mathbb{R}$ bilinear, continuous, and coercive. Let $f \in V'$, then there exists a unique $u \in V$ satisfying

$$a(u, v) = f(v) \quad \forall v \in V.$$

And the solution is stable in the sense

$$\|u\|_V \leq \frac{1}{\alpha} \|f\|_{V'}.$$

Two results needed:

(1) Riesz representation theorem: Let V be a real Hilbert space and let $f \in V'$. Then there exists a unique $g \in V$ such that

$$f(v) = (g, v)_V \quad \forall v \in V$$

Moreover: $\|g\|_V = \|f\|_{V'}$

(2) Contraction Mapping Principle: Let V be a real Hilbert space and let $T: V \rightarrow V$ be a contraction, i.e., there exists $\theta \in [0, 1]$ such that

$$\|T(v) - T(w)\|_V \leq \theta \|v - w\|_V \quad \forall v, w \in V.$$

Then there exists a unique $u \in V$ such that $T(u) = u$.

Proof sketch of Lax-Milgram (as Brenner)

↳ For each $w \in V$, consider the map

$$v \mapsto a(w, v)$$

Since a is bilinear and continuous, this map is a cont. linear form on V . By Riesz thm, there exists a unique $B(w) \in V$ such that

$$(B(w), v)_V = a(w, v) \quad \forall v \in V$$

which defines a map $B: V \rightarrow V$

↳ Notice that B is linear:

$$B(\lambda_1 w_1 + \lambda_2 w_2) = \lambda_1 B(w_1) + \lambda_2 B(w_2) \quad (*)$$

$$\begin{aligned} (B(\lambda_1 w_1 + \lambda_2 w_2), v)_V &= a(\lambda_1 w_1 + \lambda_2 w_2, v) \\ &= \lambda_1 a(w_1, v) + \lambda_2 a(w_2, v) \\ &= (\lambda_1 B(w_1) + \lambda_2 B(w_2), v)_V \end{aligned}$$

Uniqueness of Riesz rep. thm gives (*)

↳ B is bounded:

$$\|B(w)\|_V = \|a(w, \cdot)\|_{V'} = \sup_{v \in V \setminus \{0\}} \frac{|a(w, v)|}{\|v\|_V}$$

$$\uparrow_{\text{cont: } |a(w, v)| \leq f \|w\|_V \|v\|_V}$$

$$\leq f \|w\|_V.$$

$$(B(w), w)_V = a(w, w) \geq \alpha \|w\|_V^2 \quad \forall w \in V.$$

↳ Notice: $\alpha \leq \beta$

↳ Because $f \in V'$, have Riesz rep.

$$f(v) = (g, v)_V \quad \forall v \in V$$

↳ Set $0 < p < \frac{2\alpha}{\beta^2}$ and define $T: V \rightarrow V$ with

$$T(w) = w - p(B(w) - g)$$

... T is a contraction

↳ Thus there exists a unique $u \in V$ such that

$$T(u) = u$$

which means

$$u - p(B(u) - g) = u$$

and since $p > 0$, have $B(u) - g = 0$ and thus $B(u) = g$

This means

$$a(u, v) = (B(u), v)_V = (g, v)_V = f(v) \quad \forall v \in V.$$

Uniqueness of solution: say there are $u_1, u_2 \in V$

$$a(u_1 - u_2, v) = f(v) - f(v) = 0$$

using coercivity with $v = u_1 - u_2$ gives

$$\alpha \|u_1 - u_2\|_V^2 \leq a(u_1 - u_2, u_1 - u_2) = 0$$

Because $\alpha > 0$, only $u_1 = u_2$ possible.

↳ Choosing $u=v$ and coercivity gives

$$\alpha \|u\|_V^2 \leq a(u, u) = f(u) \leq \|f(u)\|$$

With dual norm

$$|f(u)| \leq \|f\|_{V'}, \|u\|_V$$

then obtain

$$\|u\|_V \leq \frac{1}{\alpha} \|f\|_{V'}$$

Interpretation: Consider $f \in V'$ and a $\tilde{f} \in V'$ with

$$a(u, v) = f(v), \quad a(\tilde{u}, v) = \tilde{f}(v) \quad \forall v \in V$$

Then $\tilde{u} = u - \tilde{u}$ satisfies

$$a(\tilde{u}, v) = f(v) - \tilde{f}(v) = \tilde{f}(v) \quad \forall v \in V$$

$$\|\tilde{u}\|_V = \|u - \tilde{u}\|_V \leq \frac{1}{\alpha} \|f - \tilde{f}\|_{V'}$$

Thus a small α indicates high sensitivity to perturbations

↳ α acts as a condition number

↳ thermal block: the smaller $\mu_{\min}$, the poorer conditioned is the problem

We are interested in uniform well-posedness and stability

$$\alpha_* = \inf_{\mu \in D} \alpha(\mu) > 0, \quad \gamma_*^* = \sup_{\mu \in D} \gamma(\mu) < \infty$$

Galerkin:

↳ We want to compute, so let us consider a finite-dimensional subspace $V_N \subset V$ of dimension $N \in \mathbb{N}$.

↳ Consider the full model

$$a(u_N(\mu), v_N(\mu)) = f(v_N(\mu)) \quad \forall v_N \in V_N$$

↳ We "trust" the full model; it achieves the accuracy we would like to achieve.

↳ Because $V_N \subset V$ subspace, coercivity and continuity are inherited from the problem formulation over V .

Theorem: Under assumptions of Lax-Milgram, for each $\mu \in D$, there exists a unique (Galerkin) solution $u_N \in V_N$ such that

$$a(u_N, v_N(\mu)) = f(v_N(\mu)) \quad \forall v_N \in V_N$$

and

$$\|u_N(\mu)\|_V \leq \frac{1}{\alpha(\mu)} \|f(\cdot, \mu)\|_{V'}.$$

Proof: Because V_N is subspace of V .

Cea's Lemma: Fix $\mu \in D$. Let $a(\cdot, \cdot; \mu)$ be cont. with constant $\gamma(\mu)$ and coercive with constant $\alpha(\mu) > 0$. Let $u(\mu) \in V$ and $u_N(\mu) \in V_N$ solve the "truth" and "full" problems, with $V_N \subset V$. Then

$$\|u(\mu) - u_N(\mu)\|_V \leq \frac{\gamma(\mu)}{\alpha(\mu)} \inf_{v_N \in V_N} \|u(\mu) - v_N\|_V$$

Proof: Let $e = u(\mu) - u_N(\mu)$. Subtract the "truth" and "full" equations shows that for all $w_N \in \underline{V_N}$

$$a(e, w_N; \mu) = 0$$

Therefore, using $e = (u - u_N) + (u_N - u_N)$

$$\begin{aligned} \alpha(\mu) \|e\|_V^2 &\leq a(e, e; \mu) \\ &= a(e, u - u_N; \mu) + a(e, u_N - u_N; \mu) \\ &= a(e, u - u_N; \mu) \\ &\leq |a(e, u - u_N; \mu)| \\ &\leq \gamma(\mu) \|e\|_V \|u - u_N\|_V \end{aligned}$$

If $e = 0$, result is immediate. Otherwise divide by $\alpha(\mu) \|e\|_V$ to obtain

$$\|u - u_N\|_V \leq \frac{\delta(\mu)}{\alpha(\mu)} \|u - v_N\|_V$$

Since this holds for all $v_N \in V_N$, we can take the infimum.

Interpretation of Cea's lemma:

$$\underbrace{\|u - u_N\|_V}_{\text{error}} \leq \underbrace{\frac{\delta(\mu)}{\alpha(\mu)}}_{\text{stability}} \underbrace{\inf_{v_N \in V_N} \|u_N - v_N\|_V}_{\text{best-approximation}}$$

Thus the Galerkin method is accurate if

- ① the approximation space contains a 'good' approximation of the exact solution
- ② the problem is stable, meaning δ/α is not 'large'.

Symmetric case: Fix $\mu \in D$. Suppose $a(\cdot, \cdot; \mu)$ is bilinear, coercive, and symmetric. Then

$$(w, v)_{a, \mu} = a(w, v; \mu)$$

is an inner product on V . Its induced norm is

$$\|v\|_{a, \mu} = \sqrt{a(v, v; \mu)}$$

if bilinear form is also continuous, then

$$\sqrt{\alpha(\mu)} \|v\|_V \leq \|v\|_{a, \mu} \leq \sqrt{\delta(\mu)} \|v\|_V$$

Prop: Define the Galerkin error

$$e_N(\mu) = u(\mu) - u_N(\mu),$$

then

$$a(e_N(\mu), v_N(\mu)) = 0 \quad \forall v_N \in V_N.$$

Proof: $\forall v_N \in V_N$, have

$$\begin{aligned} a(e_N(\mu), v_N(\mu)) &= a(u - u_N, v_N(\mu)) \\ &= a(u, v_N(\mu)) - a(u_N, v_N(\mu)) \\ &= f(v_N(\mu)) - f(v_N(\mu)) \\ &= 0 \end{aligned}$$

Theorem: Assume $a(\cdot, \cdot, \mu)$ is sym, additionally to the assn. of Cea's lemma, then

$$\|u(\mu) - u_N(\mu)\|_{0,\mu} = \inf_{v_N \in V_N} \|u(\mu) - v_N\|_{0,\mu}$$

For the $(\cdot, \cdot)_V$ inner product, obtain:

Corollary: Under assn. of Cea's lemma and sym. a , obtain

$$\|u(\mu) - u_N(\mu)\|_V \leq \sqrt{\frac{f(\mu)}{\alpha(\mu)}} \inf_{v_N \in V_N} \|u(\mu) - v_N\|_V$$

Proof: Next time